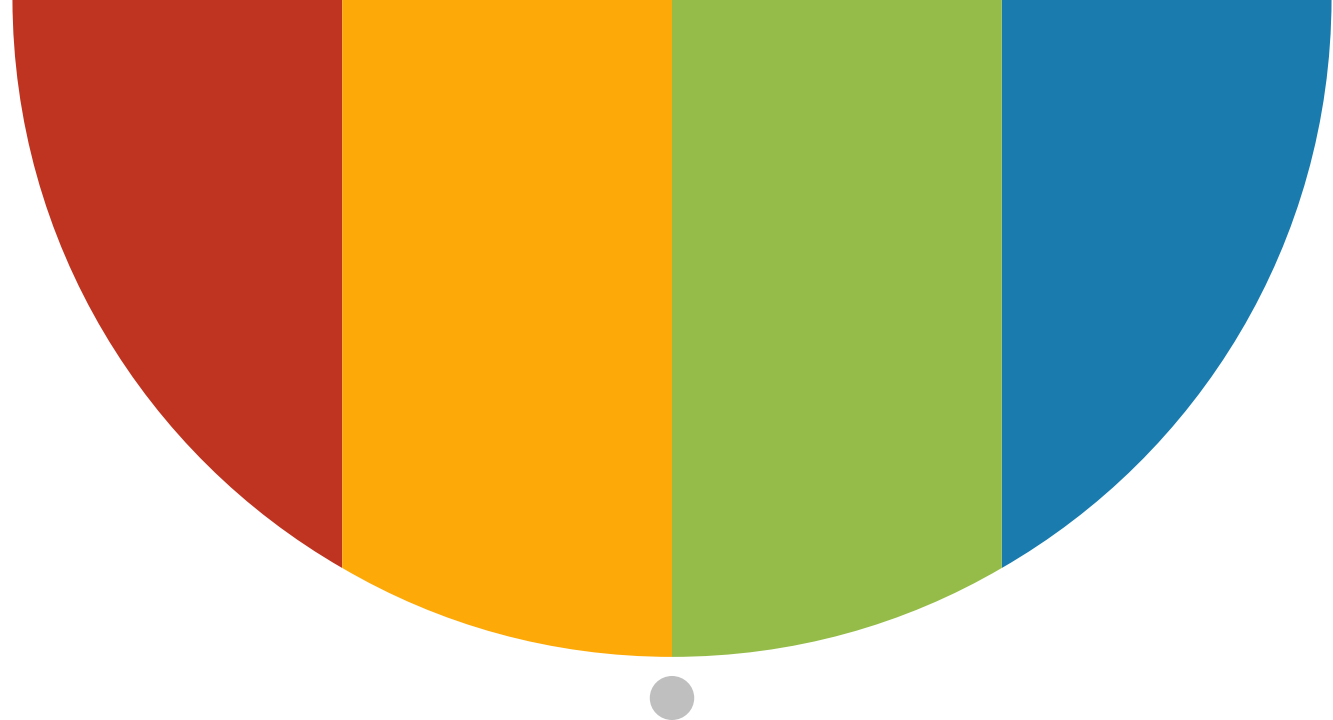




# 深度学习 Deep Learning

南开大学  
孙羽菲

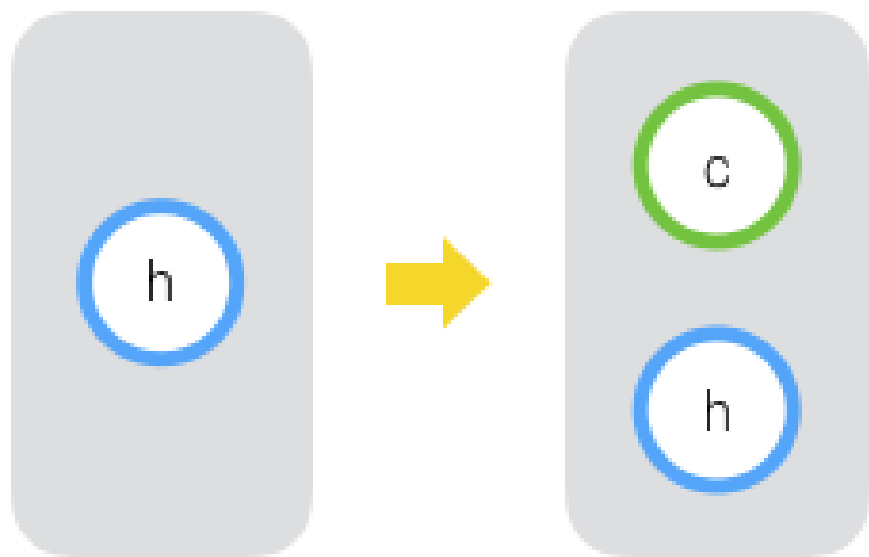
# 序列模型

RNN、LSTM

# LSTM

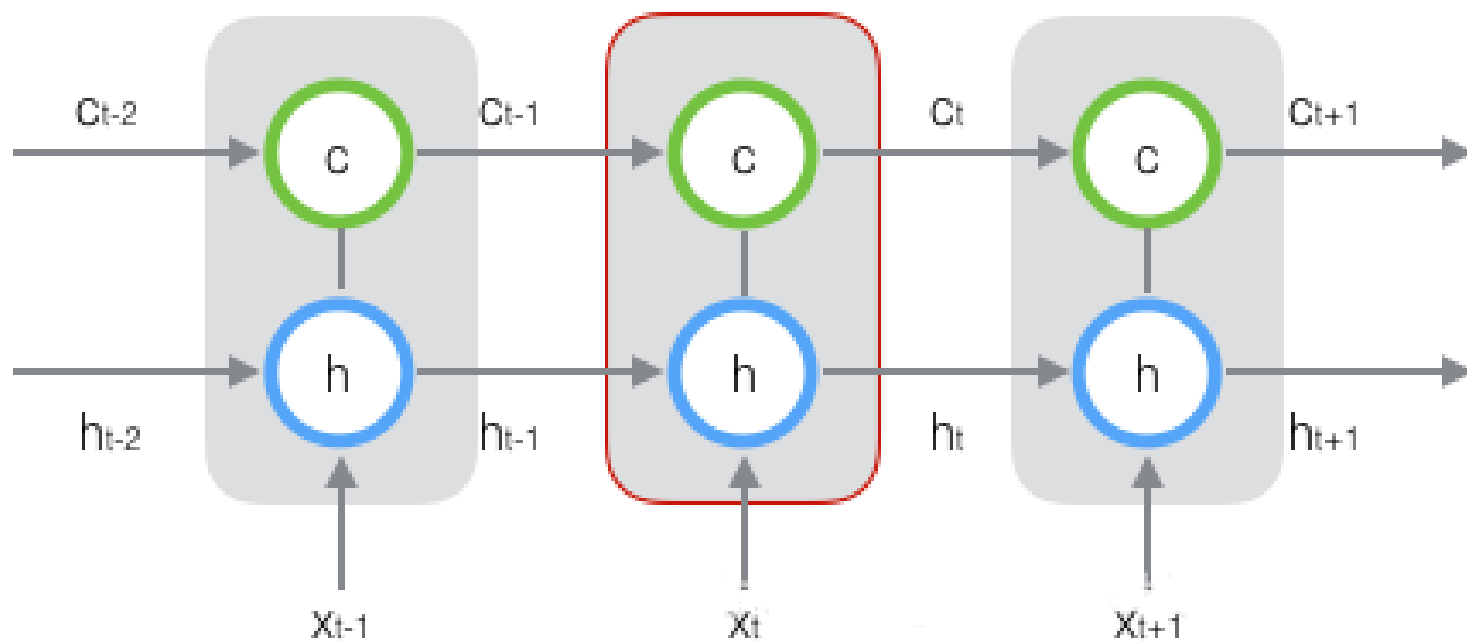
## 长短时记忆网络(Long Short Term Memory Network, **LSTM**)

单元状态(cell state)

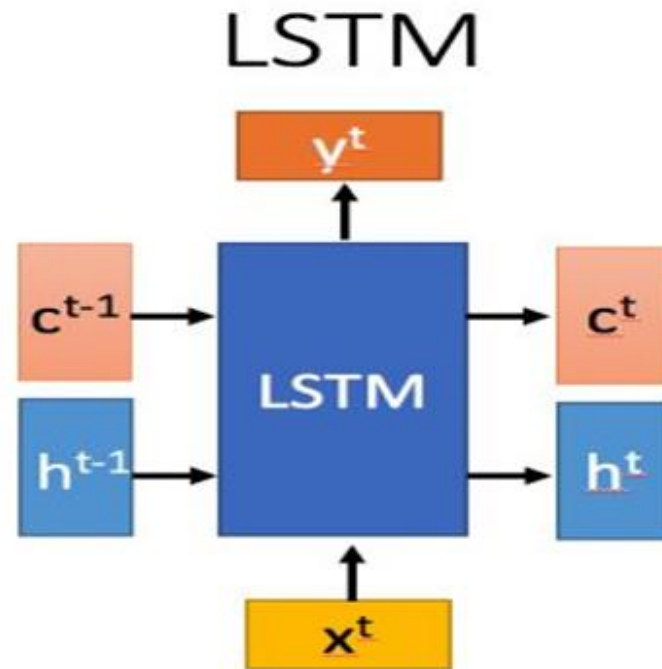
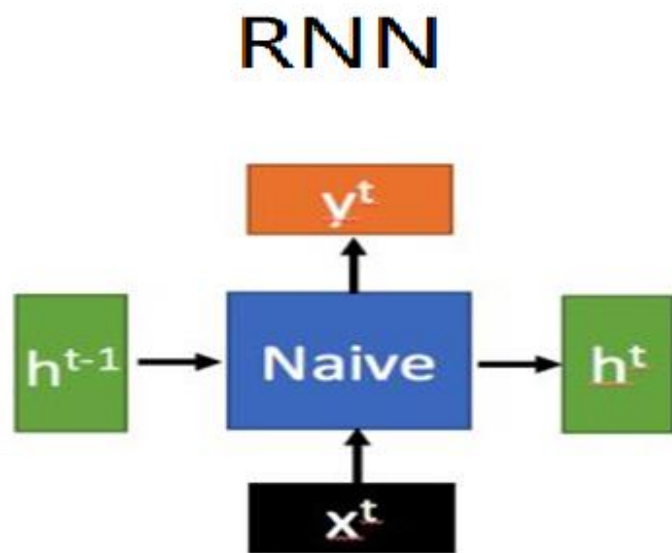


原始RNN

LSTM



# RNN & LSTM



c change slowly ➡  $c^t$  is  $c^{t-1}$  added by something

h change faster ➡  $h^t$  and  $h^{t-1}$  can be very different

RNN: 梯度爆炸或者梯度消失, 难于解决长距离依赖问题

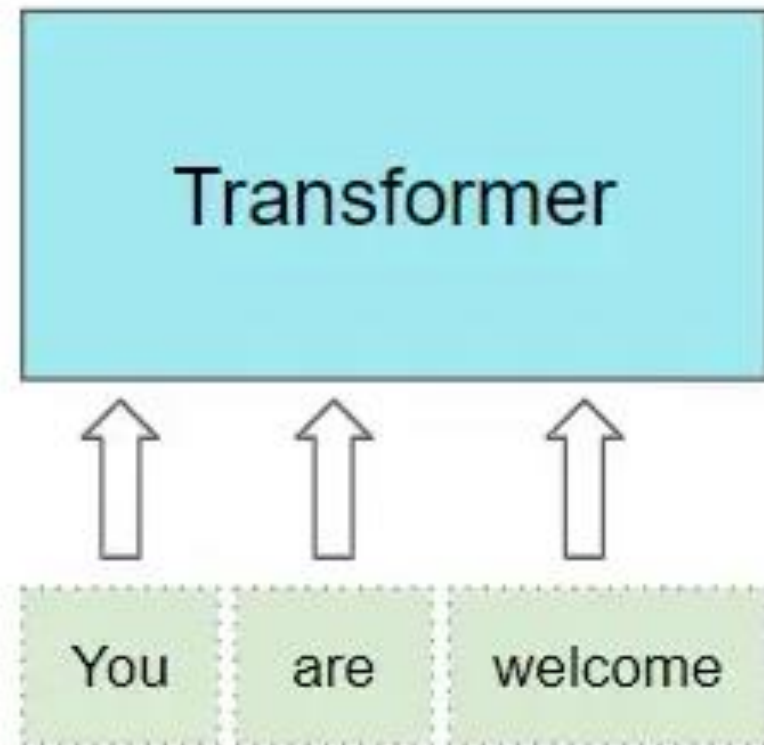
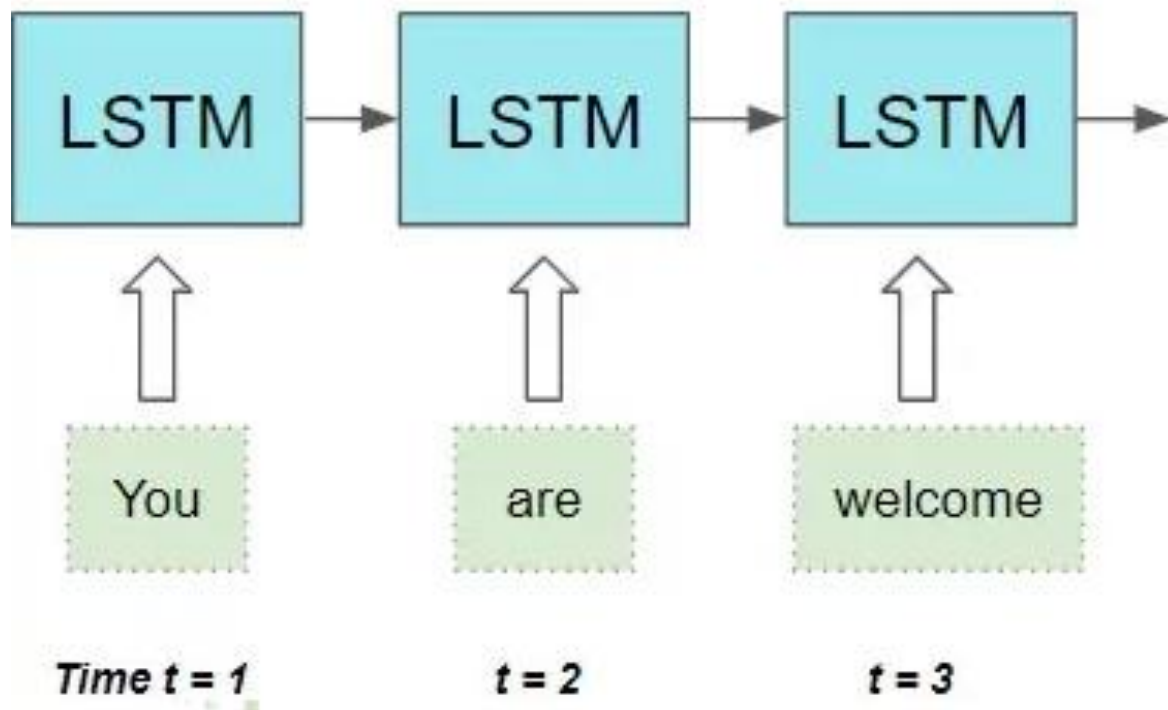
LSTM: 无法并行, 计算慢, 训练时间长

2

# Transformer

大模型技术的骨干模型

# Transformer



- ✓ 注意力机制
- ✓ 并行化训练

# Attention

Input

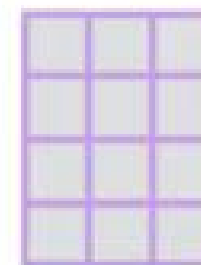
Thinking

Machines

Embedding

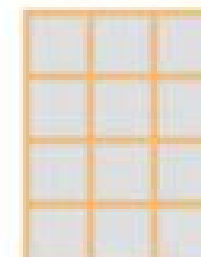


Queries



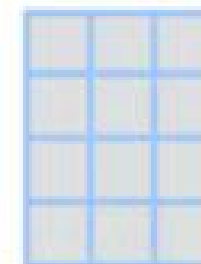
$W^Q$

Keys



$W^K$

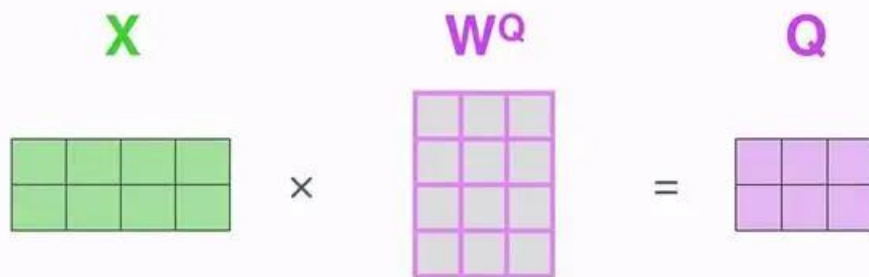
Values



$W^V$

# Self-Attention

查询矩阵Q (query)



键矩阵K (key)



值矩阵V (value)



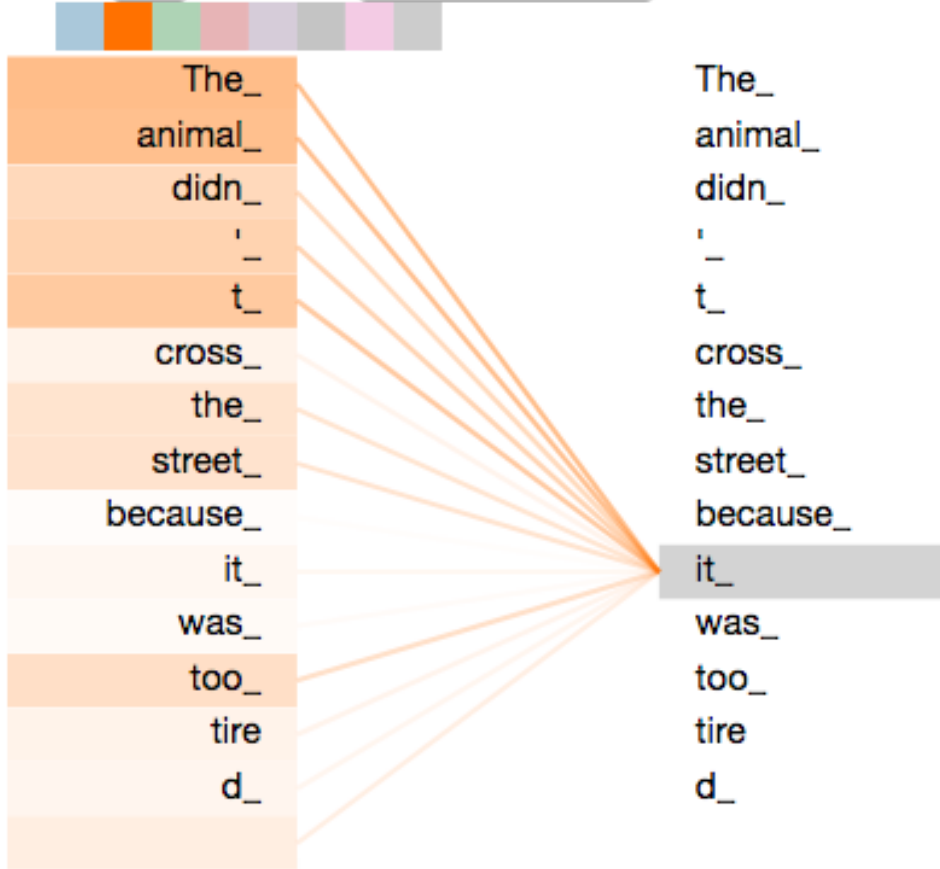
$$\text{softmax}\left(\frac{Q \times K^T}{\sqrt{d_k}}\right) \times V = Z$$



# Attention

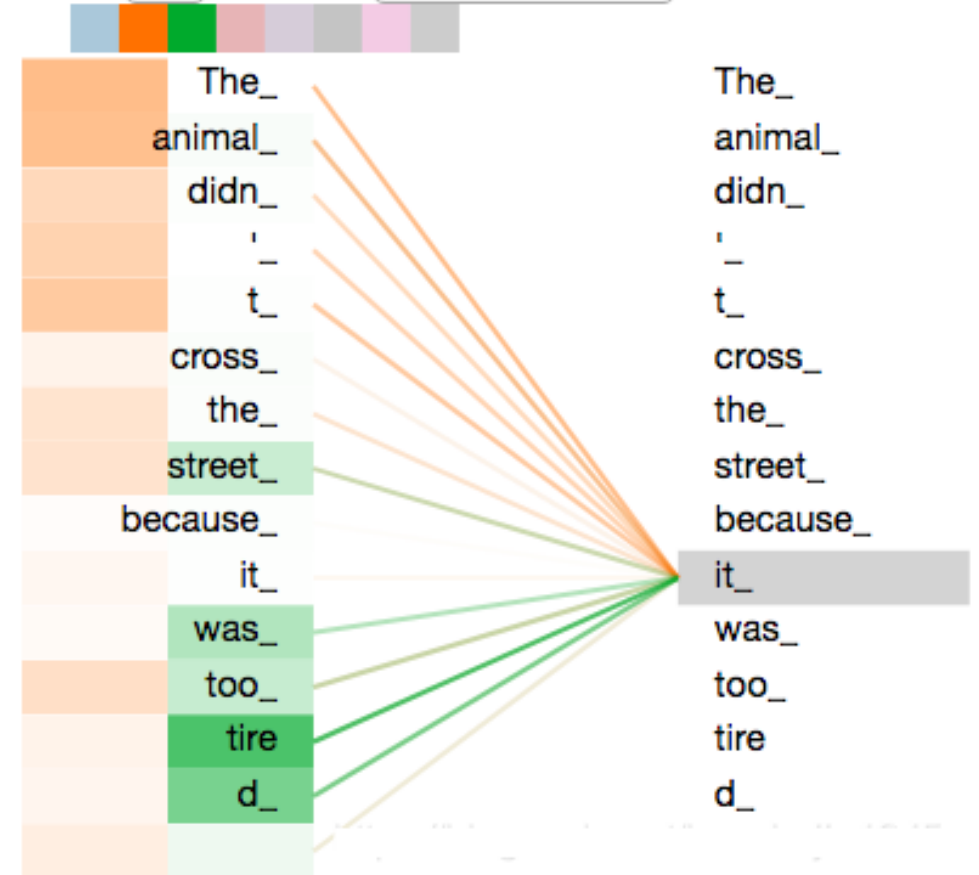
The animal didn't cross the street because **it** was too tired

Layer: 5 Attention: Input - Input



Self-Attention

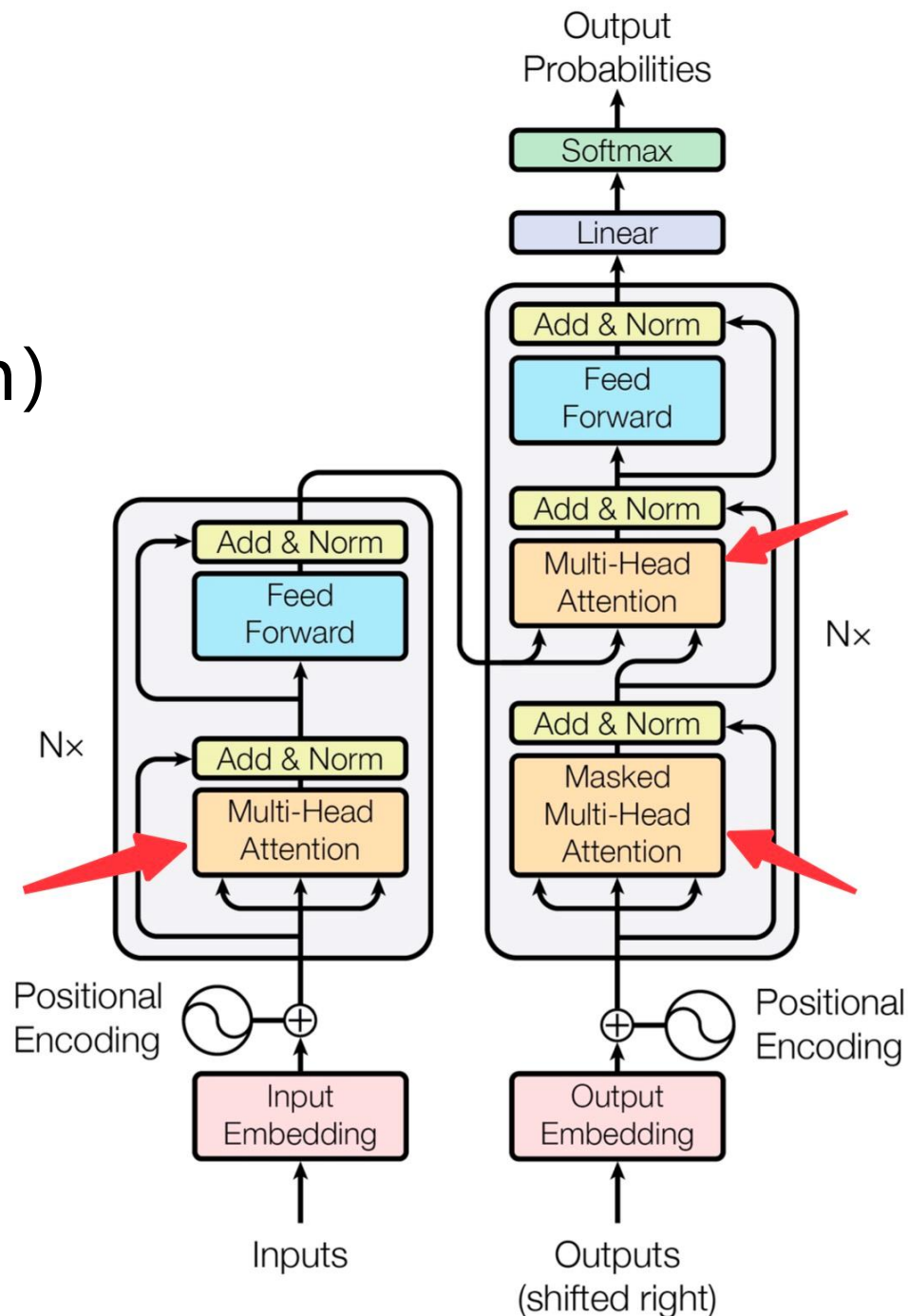
Layer: 5 Attention: Input - Input



Multi-head Attention

# Transformer

- ✓ 自注意力机制 (Self-Attention)
- ✓ 多头注意力 (Multi-Head Attention)
- ✓ 堆叠层 (Stacked Layers)
- ✓ 位置编码 (Positional Encoding)
- ✓ 残差连接和层归一化 (Residual Connections and Layer Normalization)
- ✓ 编码器和解码器 (Encoder & Decoder)





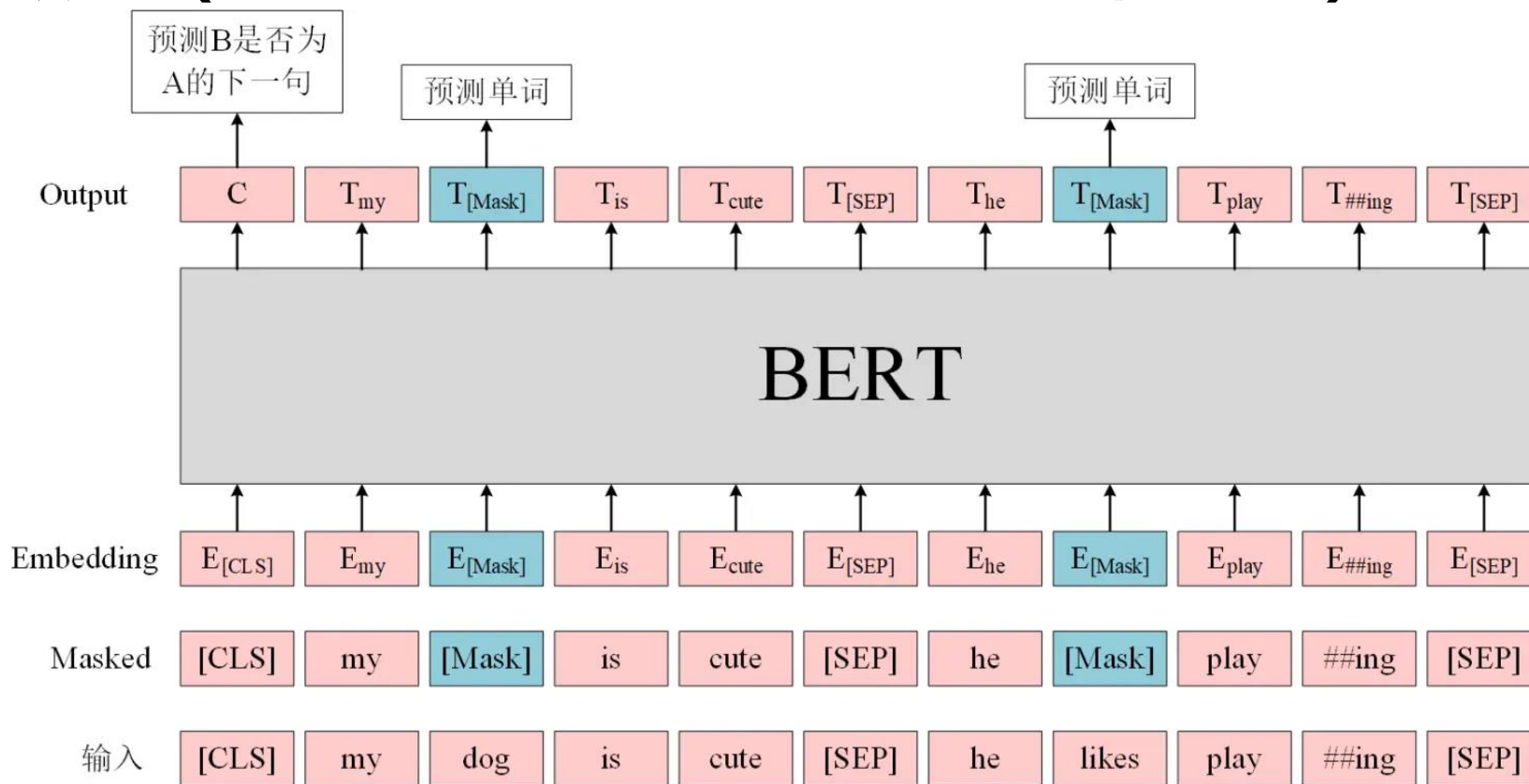
# 大语言模型

Large Language Model

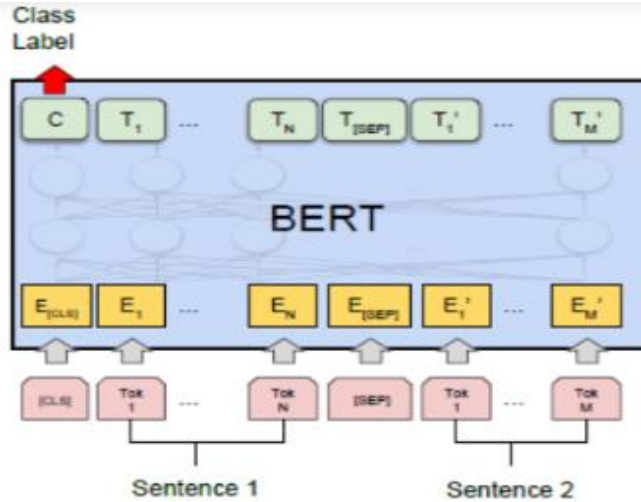
# Bert - Pretraining

**结构：**采用了 Transformer 的 Encoder 结构，BERT-base 模型包含 12 个 Encoder block，BERT-large 包含 24 个 Encoder block。

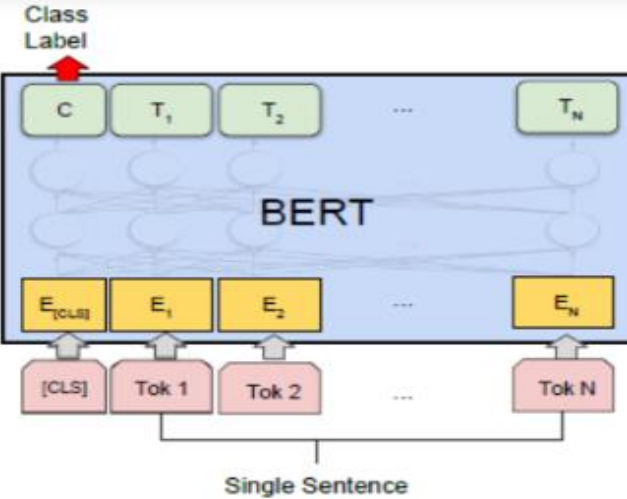
**2个预训练任务：掩码语言模型（Masked Language Model, MLM）和下一句预测（Next Sentence Prediction, NSP）。**



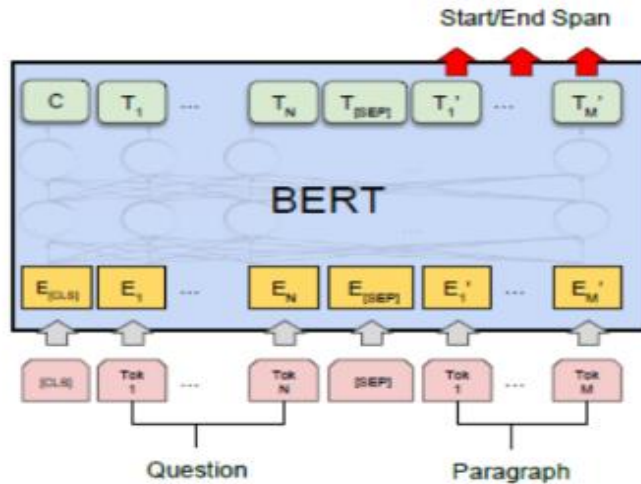
# Bert - Finetuning



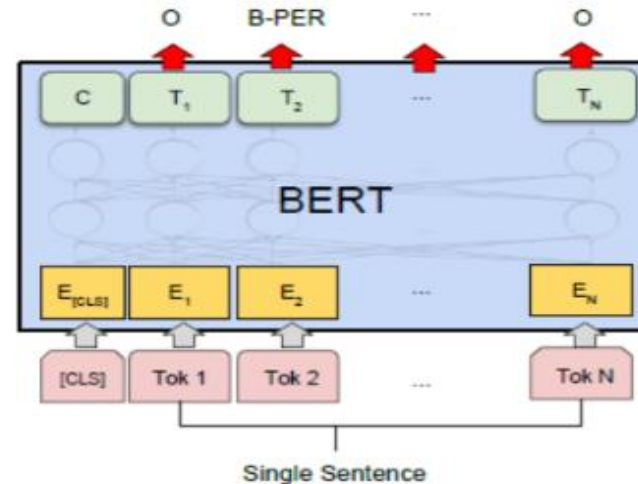
(a) Sentence Pair Classification Tasks:  
MNLI, QQP, QNLI, STS-B, MRPC,  
RTE, SWAG



(b) Single Sentence Classification Tasks:  
SST-2, CoLA



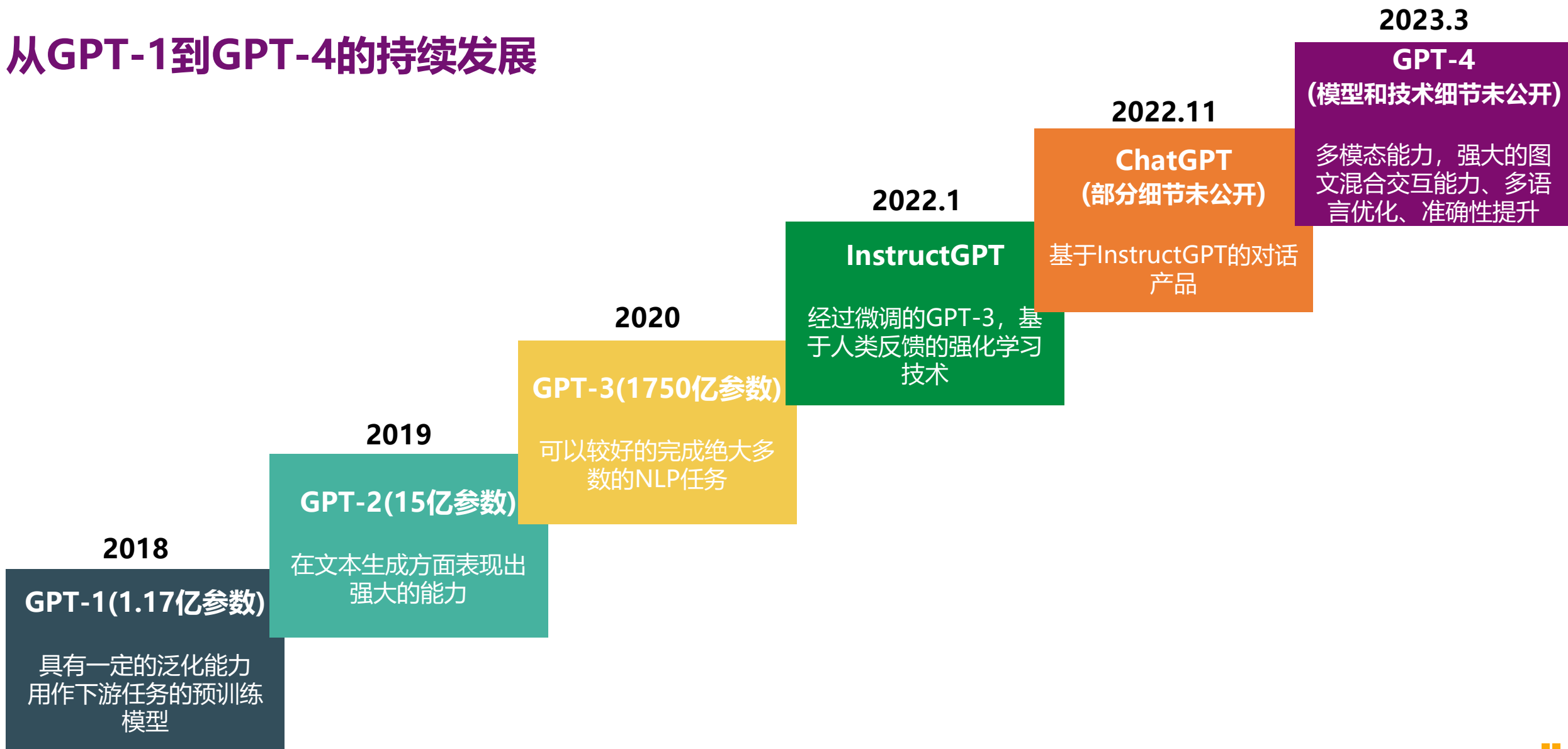
(c) Question Answering Tasks:  
SQuAD v1.1



(d) Single Sentence Tagging Tasks:  
CoNLL-2003 NER

# GPT系列发展历程

## 从GPT-1到GPT-4的持续发展

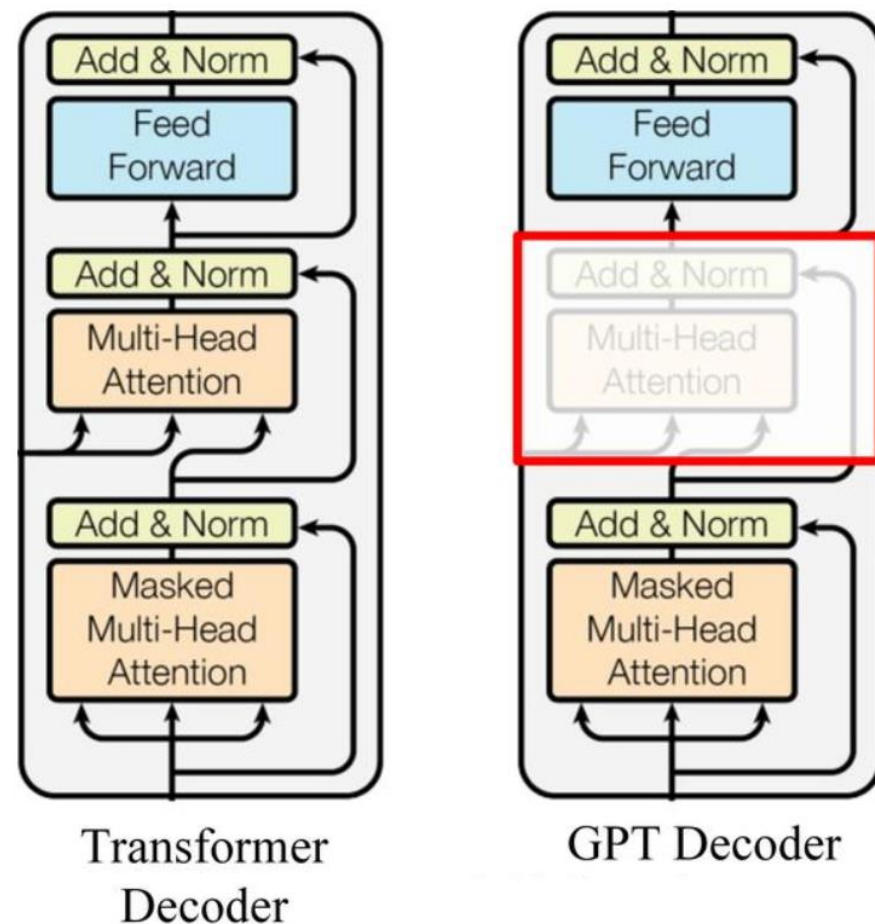
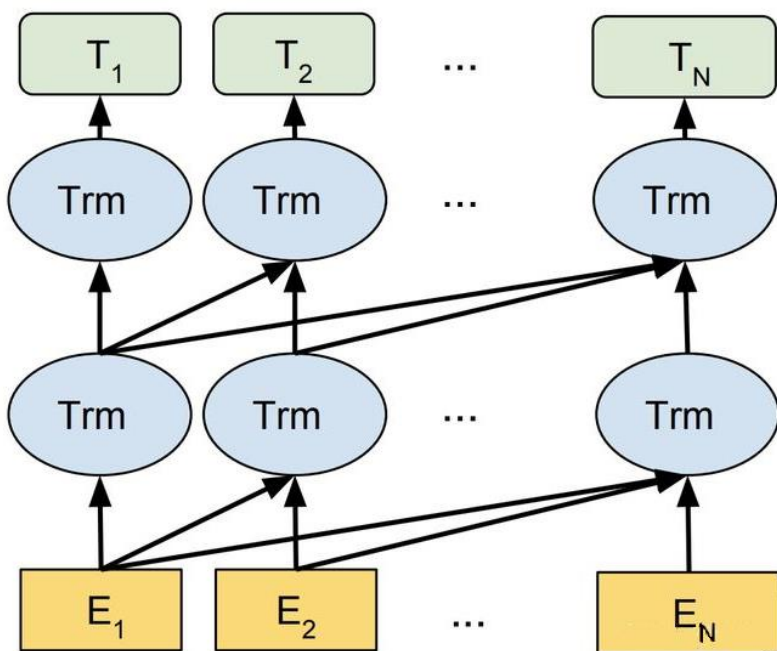


# GPT-1

**训练数据集：5GB**（BooksCorpus dataset，包含了超过7000本多领域的书。  
Word Benchmark，在句子层面进行了打乱。）

**模型结构：GPT1基于Transformer的Decoder**，12层的decoder-only transformer with masked self-attention heads。

**参数量：1.17亿**      **窗口大小：512**





# GPT-2

2019年2月 GPT-2: 多任务学习的探索

论文: 《Language Models are Unsupervised Multitask Learners》

数据集: WebText, 百万级别的文本的数据集, 约40GB

参数规模: 15亿

窗口大小: 1024

| Parameters | Layers | $d_{model}$ |
|------------|--------|-------------|
| 117M       | 12     | 768         |
| 345M       | 24     | 1024        |
| 762M       | 36     | 1280        |
| 1542M      | 48     | 1600        |



# GPT-3

**GPT-3: 2020年5月**

**数据集: 45TB (570G)**

**参数规模: 1750亿**

**模型结构: 96层**

**Sparse Attention**

**窗口大小: 2048**

**训练成本: 1200万美元**

**论文: 《Language Models are Few-Shot Learners》**

The ELMo paper? 15 pages.

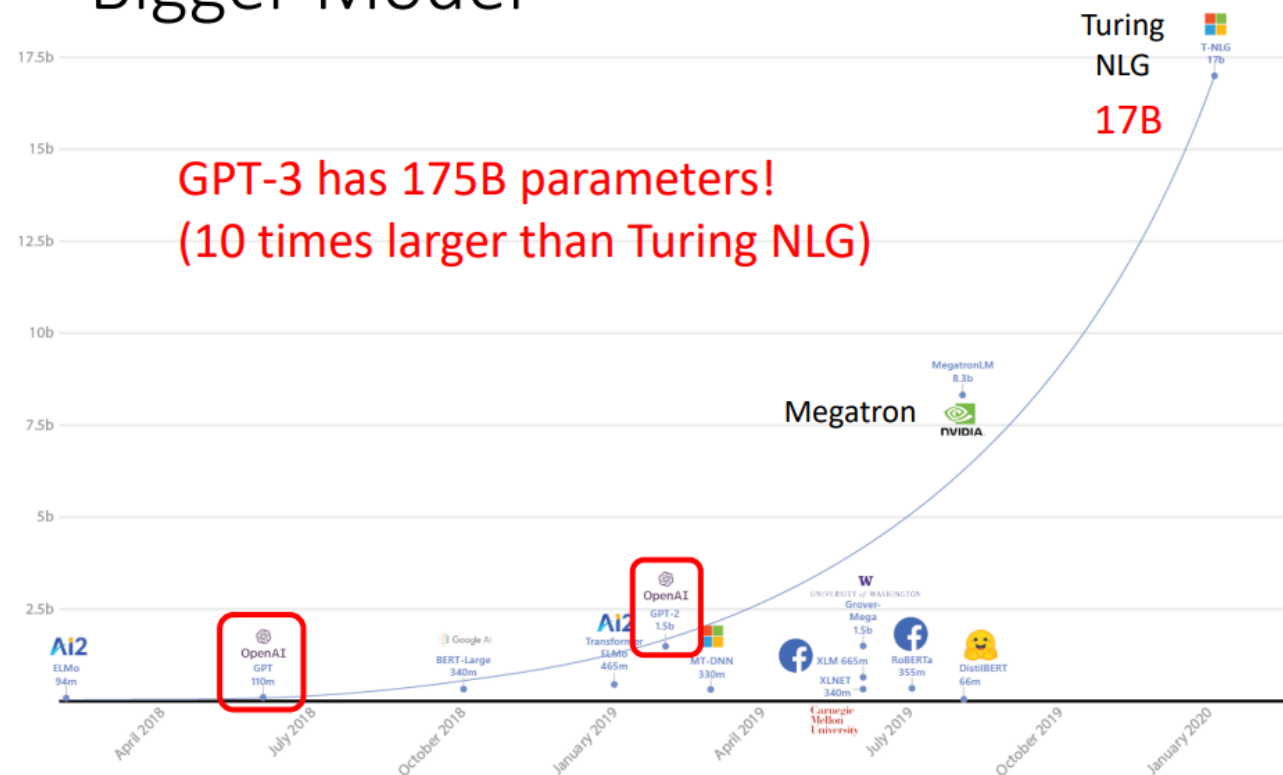
BERT? 16 pages.

GPT-2? 24 pages.

T5? 53 pages.

GPT-3?? 72 pages!

## Bigger Model



### Language Models are Few-Shot Learners

| Tom B. Brown*     |                   | Benjamin Mann*     |  | Nick Ryder*        |  | Melanie Subbiah* |  |
|-------------------|-------------------|--------------------|--|--------------------|--|------------------|--|
| Jared Kaplan†     | Prafulla Dhariwal | Arvind Neelakantan |  | Pranav Shyam       |  | Girish Sastry    |  |
| Amanda Askell     | Sandhini Agarwal  | Ariel Herbert-Voss |  | Gretchen Krueger   |  | Tom Henighan     |  |
| Rewon Child       | Aditya Ramesh     | Daniel M. Ziegler  |  | Jeffrey Wu         |  | Clemens Winter   |  |
| Christopher Hesse | Mark Chen         | Eric Sigler        |  | Mateusz Litwin     |  | Scott Gray       |  |
| Benjamin Chess    |                   | Jack Clark         |  | Christopher Berner |  |                  |  |
| Sam McCandlish    | Alec Radford      | Ilya Sutskever     |  | Dario Amodei       |  |                  |  |

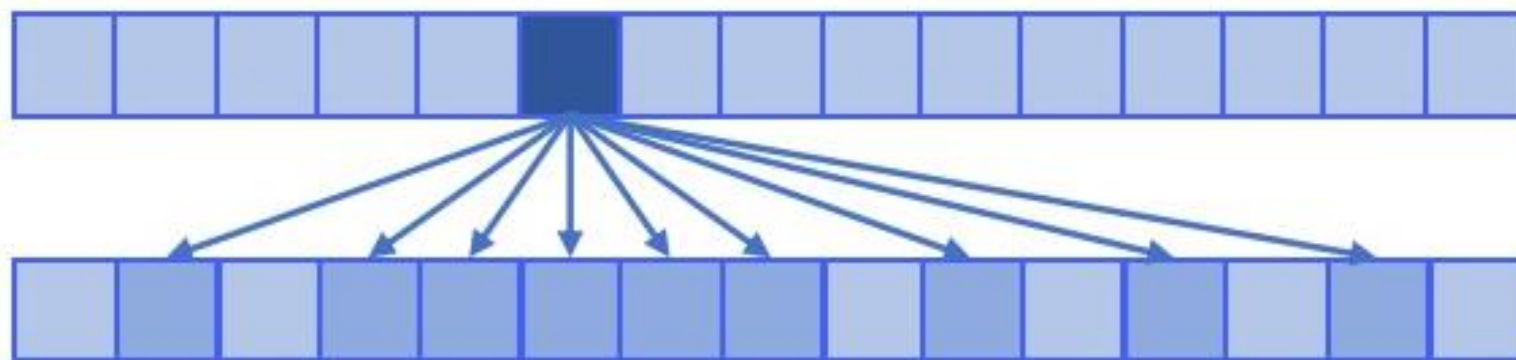
OpenAI

# GPT-3

Dense attention: 每个 token 之间两两计算 attention, 复杂度  $O(n^2)$ 。

Sparse attention: 每个 token 只与其他 token 的一个子集计算 attention, 复杂度  $O(n \cdot \log n)$

**减少注意力层的计算复杂度**, 节约显存和耗时, 从而能够处理更长的输入序列; “**局部紧密相关和远程稀疏相关**”的特性对于距离较近的上下文关注更多, 对于距离较远的上下文关注较少。



sparse attention ( $k = 2$ )

# Instruction GPT

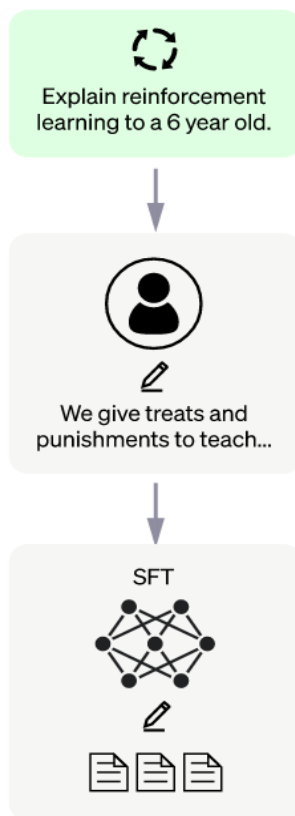
2022 年 2 月 OpenAI 发布论文《Training language models to follow instructions with human feedback》，公布 Instruction GPT 模型(**GPT-3.5**)

基于人类反馈的强化学习(RLHF, Reinforcement Learning from Human Feedback)

A prompt is sampled from our prompt dataset.

A labeler demonstrates the desired output behavior.

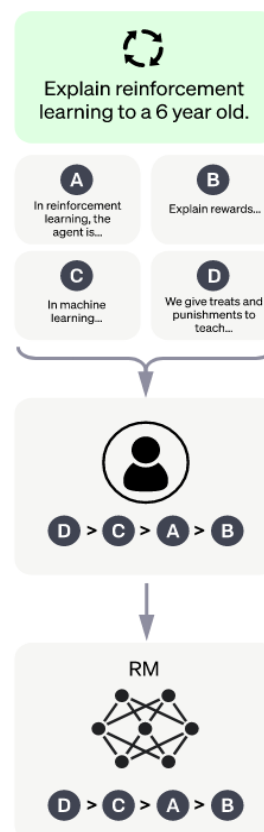
This data is used to fine-tune GPT-3.5 with supervised learning.



A prompt and several model outputs are sampled.

A labeler ranks the outputs from best to worst.

This data is used to train our reward model.



A new prompt is sampled from the dataset.

The PPO model is initialized from the supervised policy.

The policy generates an output.

The reward model calculates a reward for the output.

The reward is used to update the policy using PPO.

